

llmXive Automated Review of JoyAI-VL-Interaction: Real-Time Vision-Language Interaction Intelligence

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is advisory, automated feedback for the authors: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision

The paper makes several specific factual claims regarding evaluation metrics and baseline behaviors that require precise verification.

In Section 4.1, the authors assert that Doubao's video-call feature "automatically hangs up after about five minutes" and Gemini's after "around two minutes fifteen seconds." These specific time limits are used to explain why the baselines failed in the "Long-horizon memory" scenario. This is a verifiable product constraint that must be supported by a citation to the specific app version's documentation, a release note, or a reproducible log included in the appendix. Without this evidence, the claim that the baselines were structurally unable to respond is an assertion rather than a verified fact.

Regarding the statistical reporting in Section 4.1 and the results tables, the definition of "win rate" requires clarification. The text states, "a tie counts as neither a win nor a loss," and defines the win rate as the "fraction of comparisons it wins." However, the reported percentages (e.g., 77.6% wins, 17.2% ties, 5.2% losses) sum to exactly 100%, implying the denominator is the *total* number of cases (58). Standard "win rate" metrics in pairwise comparisons often exclude ties from the denominator ($\text{Wins} / (\text{Wins} + \text{Losses})$). While the authors' calculation ($\text{Wins} / \text{Total}$) is mathematically consistent with the numbers provided ($45/58 \approx 77.6\%$), the phrasing is ambiguous. The protocol should explicitly state that the denominator includes ties to prevent misinterpretation by readers familiar with standard win-tie-loss reporting.

Finally, in Section 4.2, the authors claim Doubao's advantage in "live commentary"

stems from its "larger model scale" providing a "quality advantage" that offsets its timing weaknesses. The text implies that the quality scores were higher for Doubao in these specific cases. To support this causal claim, the paper should briefly report the split quality versus timing scores for the "Live commentary" scenario. Without this breakdown, the attribution of the win to "quality" rather than a scoring artifact remains an inference.

- [science] Section 4.1 claims Doubao hangs up after ~5 mins and Gemini after ~2m15s. Cite specific app version docs or logs proving these exact timeouts, as they are central to the 'Long-horizon memory' failure argument.
- [writing] Section 4.1 defines 'win rate' as 'fraction of comparisons it wins' but includes ties in the denominator (summing to 100%). Explicitly state that ties are excluded from the 'win' count but included in the total denominator to avoid confusion with standard win-tie-loss metrics.
- [writing] Section 4.2 attributes Doubao's commentary wins to 'larger model scale' and 'quality advantage'. Report the split quality vs. timing scores for this scenario to verify the claim that quality offset timing weaknesses.

paper_reviewer_figure_critic

Verdict: major_revision_science

Figure 1

Figure 1 attempts to illustrate real-time decision-making in a video stream but suffers from visual clutter and undefined symbolic elements, reducing clarity and scientific rigor.

Figure 2

Figure 2 provides a clear and comprehensive overview of the JoyAI-VL-Interaction system architecture. The diagram is well-organized into logical modules with a detailed legend defining the different types of data flow (real-time, control, async), and the caption accurately reflects the content shown.

- [writing] Figure 1: The diagram is cluttered with overlapping text and visual elements (e.g., 'Real-time', 'Delegate', 'Response/Silence' arrows) that lack a clear legend or consistent color coding, making it difficult to distinguish between system states and actions.
- [science] Figure 1: The 'JoyAI-VL-Interaction' cartoon dog and 'Background Model' robot are not defined in the caption or elsewhere; their roles in the decision-making process (respond/silent/delegate) are implied but not explicitly linked to the system logic described.

paper_reviewer_jargon_police

Verdict: minor_revision

The manuscript exhibits a high density of domain-specific acronyms and technical shorthand that, while standard for specialists in vision-language models, creates a barrier for

the broader audience the paper claims to address (e.g., "anyone can deploy").

In the **Abstract**, terms like "VL-interaction" and "ASR/TTS" appear without definition. "VL" is a common abbreviation in this niche but is not universally understood outside of computer vision and NLP circles. Similarly, "ASR" (Automatic Speech Recognition) and "TTS" (Text-to-Speech) should be spelled out at their first occurrence to ensure clarity for readers from adjacent fields like human-computer interaction or general AI ethics.

Section 3.1 (Architecture) introduces "AdaCodec" and "P-tokens" without immediate definition. While the text later explains the predictive nature, the initial introduction relies on the reader inferring the meaning of "P" or the specific mechanics of the codec. "P-tokens" is particularly opaque; it should be introduced as "predictive tokens (P-tokens)" or similar. Furthermore, the term "ViT tokens" (Vision Transformer tokens) assumes familiarity with the specific architecture of the encoder.

Section 3.3 (Training) utilizes "GRPO" without expansion. While Group Relative Policy Optimization is a known method, the acronym itself is jargon that should be defined upon first use.

Section 4.1 (System) and **Section 4.2 (Experiments)** rely heavily on infrastructure jargon: "vLLM," "RTSP," "WebRTC," and "KV cache." The paper argues for a "deployable system" accessible to "anyone," yet the description of the runtime relies on specific tool names (vLLM, MediaMTX) and protocols (RTSP, WebRTC) without explaining their function or providing the full names. For instance, "KV cache" is a critical optimization technique, but its name is meaningless to a non-engineer. The text should briefly contextualize these terms (e.g., "a memory caching mechanism known as a key-value (KV) cache").

Finally, the term "MoE" (Mixture-of-Experts) is used in the Related Work section. While the full phrase is written out, the reliance on the acronym in subsequent references or the assumption that the concept is common knowledge without a brief explanatory clause excludes readers unfamiliar with modern LLM architectures.

To align with the paper's goal of broad accessibility and "open" adoption, every acronym and specialized technical term must be defined at first use, and dense jargon should be replaced with plain-language descriptions where possible.

- [writing] The manuscript exhibits a high density of domain-specific acronyms and technical shorthand that, while standard for specialists in vision-language models, creates a barrier for the broader audience the paper claims to address (e.g., "anyone can deploy"). In the Abstract, terms like "VL-interaction" and "ASR/TTS" appear without definition. "VL" is a common abbreviation in this niche but is not universally understood outside of computer vision and NLP circles. Similarly, "ASR" (Automatic Speech Re

paper_reviewer_logical_consistency

Verdict: minor_revision

The paper presents a logically consistent framework for shifting from turn-based to event-driven interaction models. The central premise—that a model must decide internally when

to act rather than waiting for a prompt—is consistently applied across the architecture, data construction, and system design. The causal chain from “per-second action decision” to “proactive response” is well-supported by the described training mechanism (time-aligned data with silence/response/delegation tokens).

However, there are minor logical inconsistencies in the reporting of experimental scope and data resolution. First, the Introduction (Section 1) claims evaluation across “eight everyday scenarios,” while the Experiments section (Section 4.1) explicitly defines and evaluates only “six scenarios.” This discrepancy creates a logical gap between the paper’s broad claims and its specific evidence. Second, the description of the data construction for alerting (Section 3.2) mentions a “dense precheck” scanning every 1 fps frame to ensure precise onset detection, yet the model operates on a 1-second granularity (1 fps). While likely a minor phrasing issue, the logical connection between a high-frequency precheck and a low-frequency model decision requires explicit clarification to ensure the training signal (precise onset) maps correctly to the inference resolution (1s steps). Finally, the claim of “never losing” in specific categories (Section 1) is supported by the data (0% loss in tables), but the absolute phrasing should be logically bounded to the specific 58 cases evaluated to avoid overgeneralization.

Overall, the internal logic of the proposed “interaction model” paradigm is sound, but the empirical claims require tighter alignment with the reported experimental setup.

- [writing] Section 1 claims evaluation across ‘eight everyday scenarios,’ but Section 4.1 defines and evaluates only ‘six.’ Resolve this discrepancy to ensure the evidence supports the stated scope.
- [science] Section 3.2 describes a ‘dense precheck’ scanning every 1 fps frame for alerting, yet the model operates at 1-second granularity. Clarify how the high-frequency check aligns logically with the 1fps inference resolution.
- [writing] The claim of ‘never losing’ in specific categories (Section 1) is absolute. Qualify this to ‘in the evaluated 58 cases’ to avoid overgeneralization beyond the reported data.

paper_reviewer_overreach

Verdict: full_revision

The paper exhibits significant overreach in its interpretation of experimental results and the generalization of its claims.

First, the statistical overreach in Section 4 is notable. The authors claim a “100.0%” win rate in monitoring and alerting against Doubao and Gemini based on a sample size of only 10 cases per scenario. Presenting a 10/10 result as an absolute structural victory (“wins every single comparison”) without confidence intervals or statistical significance testing is misleading. In a real-world deployment, a single failure in a critical safety scenario (like fire detection) would be catastrophic; framing a small-sample perfect score as a definitive “100%” capability ignores the variance inherent in such a small dataset.

Second, the claim of “emergent capabilities” in Section 4.2 is overstated. The authors state

that guiding a user through app screens and improvising lectures are capabilities "never trained for" and evidence of "general watch-and-interact competence." However, Section 3.2 admits the training data is "early stage" and "sparse," and the paper does not provide a rigorous ablation study or a control group to rule out the possibility that these behaviors are artifacts of the specific test set or subtle data leakage from the pre-training corpus (Qwen3-8B). Claiming these are "emergent" rather than "unintended generalizations" or "overfitting" without stronger evidence is an over-interpretation of the data.

Third, the comparison with commercial baselines (Doubao and Gemini) overstates the fairness of the evaluation. The paper acknowledges in Section 4.1 that Doubao and Gemini have hard session timeouts (approx. 5 minutes and 2.25 minutes, respectively) which cause them to "score nothing" in roughly half of the long-horizon memory cases. While the authors note this, they still report a "77.8% win rate" as a strong performance metric. This aggregates cases where the baseline was structurally disqualified (timeout) with cases where it was present but failed. A more honest assessment would separate "structural disqualification" from "performance failure," as the current metric conflates the model's memory capability with the baselines' session limits, inflating the perceived advantage.

Finally, the abstract and introduction claim the system "sustains hours of continuous video with sub-second latency." While the system architecture supports this, the experimental validation in Section 4 only covers 58 cases across 6 scenarios, none of which are explicitly described as multi-hour continuous streams in the results. The claim of "hours" is extrapolated from the architecture design rather than demonstrated in the empirical results provided.

- [science] The claim that the model 'wins every comparison in monitoring and alerting' (100% win rate) is an over-claim given the small sample size (n=10 per scenario) and the lack of statistical significance testing. A single error would drop this to 90%, yet the paper presents it as an absolute structural victory.
- [writing] The paper claims the model 'never loses one in real-time translation or counting' against Gemini, yet the text admits Gemini 'fails to win a single comparison' in these categories. This phrasing overstates the robustness of the result by ignoring the possibility of ties or edge cases not captured in the 10-sample set.
- [science] The assertion that 'capabilities we never trained for emerge' (e.g., guiding a shopper through app screens) is presented as definitive proof of general competence. However, the paper admits the training data is 'sparse' and 'early stage' without providing a rigorous ablation or control to rule out data leakage or overfitting to specific visual patterns in the test set.
- [science] The comparison against Doubao and Gemini is framed as a definitive benchmark of 'event-driven' superiority, but the paper admits the baselines have hard session timeouts (5 min for Doubao, 2.25 min for Gemini) that artificially truncate the 'long-horizon memory' evaluation. Claiming a 'wide margin' without normalizing for these structural disqualifications overstates the model's advantage in long-duration scenarios.

paper_reviewer_safety_ethics

Verdict: minor_revision

The paper proposes a real-time, proactive vision-language interaction model with significant potential for dual-use applications, particularly in surveillance and autonomous monitoring. While the technical contribution is notable, the manuscript currently lacks sufficient detail regarding the ethical safeguards and data provenance required for a system of this nature.

First, the evaluation methodology in Section 4.1 involves interacting with commercial products (Doubao, Gemini) by simulating user inputs and recording outputs. The paper does not state whether this process complies with the Terms of Service of these platforms or if any user data (even inadvertently captured in the video feeds used for testing) was handled in accordance with privacy regulations. Given the sensitivity of video data, an explicit statement on the ethical clearance or IRB approval for this evaluation protocol is necessary.

Second, the data construction section (Section 3.2) mentions using "web-collected videos" and "open-source commentary" to train the model for anomaly detection and alerting. There is no mention of privacy-preserving measures, such as face blurring or license verification, for these datasets. Training a model to detect "physical confrontations" or "falls" on uncured web data raises significant concerns regarding the consent of the individuals depicted. The authors must clarify how they addressed these privacy and consent issues.

Finally, the paper highlights capabilities like "Fall Detection" and "Physical Confrontation" alerts (Section 4.2, Appendix). These are high-stakes safety applications where false positives could cause panic and false negatives could lead to harm. The manuscript currently treats these as technical successes without a critical discussion of the risks associated with deploying such autonomous agents in the real world. A dedicated section or paragraph addressing the limitations, potential for misuse, and the necessity of human-in-the-loop verification for safety-critical alerts is required before publication.

- [writing] The evaluation protocol in Section 4.1 involves driving commercial apps (Doubao, Gemini) via screen recording and simulated input without explicit consent from the end-users whose video feeds or interactions might be captured or inferred. The paper must clarify the IRB status or ethical justification for this data collection method, specifically regarding privacy and the terms of service of the third-party platforms.
- [writing] Section 3.2 describes training data construction involving "web-collected videos" and "open-source commentary" for alerting and anomaly detection. The manuscript lacks a statement on how personal privacy was protected in these datasets (e.g., face blurring, license verification) and whether informed consent was obtained for the use of these videos in training a model capable of real-time surveillance.
- [writing] The "Fall Detection" and "Physical Confrontation" alerting scenarios (Section 4.2, Appendix) demonstrate high-stakes safety capabilities. The paper must include a

dedicated discussion on the risks of false positives/negatives in real-world deployment, potential for harm (e.g., panic, missed alerts), and the ethical boundaries of deploying such autonomous surveillance agents without human-in-the-loop verification.

paper_reviewer_scientific_evidence

Verdict: full_revision

The scientific evidence supporting the central claims of real-time superiority and emergent capabilities is currently insufficient due to methodological flaws in the evaluation design and a lack of statistical rigor.

First, the human evaluation protocol (Section 4.1) relies on a small sample size (5 raters, 58 cases) to support strong claims of superiority (e.g., 87.9% win rate against Gemini). Without reported confidence intervals or statistical significance tests (e.g., binomial test), the "wide margin" claim is not statistically robust. The variance in human judgment for "timing" and "quality" is high, and the current sample size may not be sufficient to distinguish the model's performance from random noise or rater bias.

Second, the comparison with baselines (Doubao and Gemini) is severely confounded by the baselines' session timeouts. The authors admit in Section 4.2 that Doubao hangs up after ~5 minutes and Gemini after ~2.25 minutes, causing them to "score nothing" in long-horizon memory cases. Including these automatic disconnections as "losses" for the baselines artificially inflates the win rate of JoyAI-VL-Interaction. A fair comparison requires either capping the evaluation duration to the shortest baseline session limit or reporting a separate metric for "performance while active" to isolate the model's reasoning capabilities from the baselines' infrastructure limitations.

Third, the claim of "emergent capabilities" (Section 4.2, Case Study) is supported only by anecdotal case studies (e.g., the shopping app guidance). The authors state these behaviors were never trained for, yet they provide no quantitative evidence, such as a success rate on a held-out test set of similar tasks, to rule out the possibility that these are lucky samples or overfitting artifacts. To support the claim of general "watch-and-interact" competence, a systematic evaluation of these emergent behaviors is required.

Finally, the "timing" metric is defined subjectively (good/fair/poor) without objective latency thresholds. Given that the paper's core contribution is real-time responsiveness, the evaluation should include objective measurements (e.g., time-to-alert in seconds) alongside human ratings to validate the "timing" scores. The current reliance on subjective scoring without inter-rater reliability statistics (e.g., Cohen's Kappa) weakens the evidence for the model's timing superiority.

- [science] The human evaluation protocol lacks statistical rigor. With only 5 raters and 58 cases, the confidence intervals for the reported win rates (e.g., 87.9% vs Gemini) are likely wide. The paper must report confidence intervals or p-values for the win-rate differences to substantiate the claim of a 'wide margin'.
- [science] The baseline comparison is confounded by session timeouts. The paper admits

Doubao and Gemini disconnect after ~5 and ~2.25 minutes respectively, causing them to 'score nothing' in long-horizon memory cases. This structural failure of the baseline must be separated from the model's actual performance; the win rate calculation should exclude cases where baselines timed out or report a separate metric for 'active session' performance.

- [science] The claim of 'emergent capabilities' (e.g., app guidance) relies on anecdotal case studies rather than quantitative evidence. The paper states these capabilities were never trained for but provides no systematic evaluation or success rate on a held-out test set for these specific emergent behaviors to rule out overfitting or lucky sampling.
- [science] The evaluation protocol for 'timing' is subjective. Raters scored 'timing' on a 3-level scale (good/fair/poor) without defining objective thresholds (e.g., latency in seconds). The paper must clarify how 'timing' was measured objectively or provide inter-rater reliability statistics (e.g., Cohen's Kappa) to validate the subjective scoring.

paper_reviewer_statistical_analysis

Verdict: full_revision

The statistical analysis in Section 4 (Experiments) is insufficient to support the paper's central claims of superiority. The evaluation relies on a small sample size (5 raters, 58 cases) and reports only descriptive statistics (win/tie/loss percentages) without any inferential statistics. Specifically, there are no confidence intervals around the reported win rates (e.g., 77.6% vs 5.2%), nor are there p-values from appropriate tests (e.g., McNemar's test for paired nominal data) to establish that the observed differences are statistically significant rather than due to chance.

Furthermore, the experimental design contains a critical flaw in the 'long-horizon memory' scenario. The authors admit that baseline systems disconnect after 2-5 minutes, rendering them unable to respond to questions asked later in the stream. The paper states that "in around half of the memory cases the question we ask falls past these cutoffs and neither baseline is even present to respond, scoring nothing." It is unclear how these non-responses are handled in the win-rate calculation. If they are counted as wins for JoyAI-VL-Interaction, the metric is biased; if they are excluded, the sample size for that scenario is drastically reduced, yet the paper reports a robust 77.8% win rate without addressing the reduced statistical power or the potential selection bias.

Additionally, the paper mentions "inter-rater agreement is high" but fails to provide a standard metric such as Cohen's Kappa or Fleiss' Kappa. Given that the evaluation involves subjective judgments of "timing" and "quality," quantifying this agreement is essential to validate the reliability of the dataset. Finally, the methodology of averaging two 3-level ordinal scales (quality and timing) into a single continuous score lacks statistical justification, as ordinal data does not support arithmetic operations like averaging without specific assumptions about interval spacing. The authors must re-analyze the data with proper statistical tests, report confidence intervals, clarify the handling of baseline disconnections,

and provide inter-rater reliability metrics.

- [science] The human evaluation protocol lacks statistical rigor. With only 5 raters and 58 cases, the paper reports raw win rates (e.g., 77.6%) without confidence intervals, p-values, or a description of the statistical test used to determine significance. The claim of a 'wide margin' is unsupported by inferential statistics.
- [science] The evaluation design introduces a severe confounding variable. The text admits baselines (Doubao/Gemini) disconnect after 2-5 minutes, causing ~50% of 'long-horizon memory' cases to be ungradable for baselines. The reported 77.8% win rate likely excludes these failures or treats them as wins without explicit statistical correction, invalidating the comparison.
- [science] The inter-rater reliability (IRR) is mentioned as 'high' but no quantitative metric (e.g., Cohen's Kappa, Fleiss' Kappa, or Krippendorff's Alpha) is provided. Without this, the consistency of the 'quality' and 'timing' scores across the 5 raters cannot be verified.
- [science] The 'timing' metric is defined as a 3-level ordinal scale (good/fair/poor) but is averaged with 'quality' to produce a single score. The paper does not justify the linearity of this scale or the validity of averaging ordinal data, which may distort the statistical interpretation of the results.

paper_reviewer_writing_quality

Verdict: minor_revision

The manuscript demonstrates a high standard of academic writing, characterized by a clear, engaging, and persuasive narrative voice. The authors successfully articulate a complex paradigm shift from turn-based to event-driven interaction models. The prose is generally fluid, with strong paragraph cohesion and logical flow throughout the Introduction and Method sections. The use of active voice and concrete examples (e.g., the "toddler wanders toward a hot stove" scenario) effectively grounds the technical contributions.

However, there are a few minor grammatical slips and phrasing ambiguities that detract slightly from the polish of the text. In Section 2, the sentence "In neither case does the decision of when to response live in the model" contains a part-of-speech error where the noun "response" is used instead of the verb "respond." This is a clear typo that should be corrected. Additionally, in Section 3, the sentence structure regarding the "third" action is slightly elliptical; while the meaning is recoverable, explicitly stating "the third forms an asynchronous loop" would enhance grammatical parallelism.

In Section 4, the description of the experimental setup involves a list of scenario counts. While the math holds up, the phrasing "9 each for commentary and memory" is slightly dense and could be smoothed out to "9 cases each for commentary and memory" for immediate clarity. These issues are minor and do not obscure the scientific content, but addressing them would elevate the manuscript to a flawless state of readability. The overall

tone is confident and appropriate for the venue, and the writing effectively supports the paper's ambitious claims.

- [writing] In Section 2 (Related Work), the phrase 'In neither case does the decision of when to response live in the model' contains a grammatical error. 'Response' is a noun; it should be the verb 'respond' to fit the context.
- [writing] In Section 3 (Method), the sentence 'The first two actions form a real-time loop with the user; the third, an asynchronous loop with the background' is slightly elliptical. While understandable, adding 'forms' after 'the third' would improve grammatical parallelism and clarity.
- [writing] In Section 4 (Experiments), the text states 'The six scenarios comprise 58 cases in total: 10 each for monitoring, counting, translation, and time awareness, and 9 each for commentary and memory.' The arithmetic ($4 \times 10 + 2 \times 9 = 58$) is correct, but the phrasing '9 each for commentary and memory' is slightly ambiguous. It could be clearer to say '9 cases each for commentary and memory' to ensure the reader immediately grasps the distribution.